Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________ 12. ________

Warm-Up: Shifts

Example (Warm-Up: Shifts):
$y = x^2 - 6$ shifts $y = x^2$ down 6 units; vertex $(0, -6)$.

Directions: Describe the shift and state the vertex for each function.

1. $y = x^2 + 4$

2. $y = (x - 3)^2$

3. $y = (x + 5)^2$

4. $y = x^2 - 2$

Reflections and Stretches

Example (Reflections and Stretches):
$y = -3x^2$ reflects $y = x^2$ over the x-axis and stretches it (narrower) by a factor of 3.

Directions: Describe the reflection and/or stretch/compression for each function.

5. $y = -x^2$

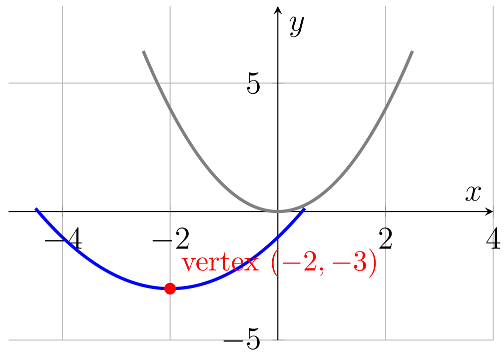
6. $y = 4x^2$

7. $y = \frac{1}{4}x^2$

8. $y = -\frac{1}{2}x^2$

Combined Transformations

Example (Combined Transformations):
$y = -2(x - 1)^2 + 4$: shift right 1, reflect over x-axis, stretch narrower by 2, shift up 4; vertex $(1, 4)$.



$y = x^2$ (gray) transformed into $y = \frac{1}{2}(x+2)^2 - 3$ (blue).

Directions: Describe all transformations (shift, reflection, stretch/compression) and state the final vertex.

9. $y = \frac{1}{2}(x+2)^2 - 3$ (use the graph above to confirm your description)

10. $y = 3(x-4)^2 - 1$

11. $y = -\dfrac{1}{3}(x+2)^2 + 5$

Challenge (same sheet):

12. Describe the transformations, in order, that turn $y = x^2$ into $y = -4(x+3)^2 - 7$.

Answer Key — fully worked

Procedure: Start from the parent function $y = x^2$.; Identify h: shift left if the equation has $(x+h)$, right if $(x-h)$.; Identify k: shift up if positive, down if negative.; Check the sign of a: negative means reflect over the x-axis.; Check the size of $|a|$: greater than 1 means narrower (stretch), between 0 and 1 means wider (compression).; Combine all transformations to state the final vertex and shape..

1. Shifts up 4; vertex $(0, 4)$.

2. Shifts right 3; vertex $(3, 0)$.

3. Shifts left 5; vertex $(-5, 0)$.

4. Shifts down 2; vertex $(0, -2)$.

5. Reflects over the x-axis; no stretch (a=-1).

6. Stretches narrower by a factor of 4; no reflection.

7. Compresses wider by a factor of $\frac{1}{4}$; no reflection.

8. Reflects over the x-axis and compresses wider by a factor of $\frac{1}{2}$.

9. Shift left 2, compress wider by $\frac{1}{2}$ (no reflection), shift down 3; vertex $(-2, -3)$ – matches the graph shown.

10. Shift right 4, stretch narrower by 3 (no reflection), shift down 1; vertex $(4, -1)$.

11. Shift left 2, reflect over the x-axis and compress wider by $\frac{1}{3}$, shift up 5; vertex $(-2, 5)$.

12. In order from the parent $y = x^2$: (1) $(x + 3)$ shifts the graph left 3 units; (2) the negative sign reflects it over the x-axis; (3) $|a| = 4 > 1$ stretches it narrower by a factor of 4; (4) the -7 shifts it down 7 units. Final vertex: $(-3, -7)$, opening down.